



2026 Course 3 - Grade 8 Mathematics

Practice Test



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x	y
-1	$-\frac{3}{2}$
$-\frac{1}{2}$	0
0	$\frac{3}{2}$
$\frac{1}{2}$	3

What are the slope and the y -intercept of the linear function that is represented by the table?

The slope is 3, and the y -intercept is $-\frac{1}{2}$.

The slope is -3, and the y -intercept is $\frac{3}{2}$.



The slope is 3, and the y -intercept is $\frac{3}{2}$.



The slope is -3, and the y -intercept is $-\frac{1}{2}$.



Practice Test



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What is the y -intercept of a line that has a slope of -3 and passes through point $(-5, 4)$?

7

-11



19

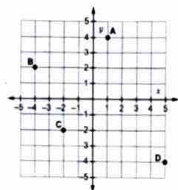
-17

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Which equation represents the line that passes through points B and C on the graph?



$y = 2x - 6$



$y = 2x + 10$

$y = -2x - 6$



$y = -2x + 10$

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Which equation represents the line that passes through points (0, 6) and (2, 0)?

$$y = -3x + 2$$

$$y = -3x + 6$$



$$y = -\frac{1}{3}x + 2$$

$$y = -\frac{1}{3}x + 6$$

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A line has a slope of $-\frac{1}{4}$ and passes through point $(-\frac{5}{4}, 1)$. What is the equation of the line?

$y = -\frac{1}{4}x + \frac{21}{16}$

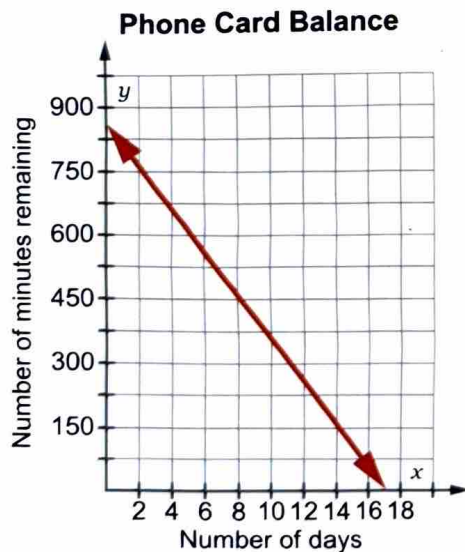
$y = -\frac{1}{4}x + \frac{11}{16}$ ✓

$y = -\frac{1}{4}x - \frac{5}{4}$

$y = -\frac{1}{4}x - \frac{1}{4}$

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Practice Test

Inez has a phone card. The graph shows the number of minutes that remain on her phone card after a certain number of days.



The slope of the line that represents the data is -50 , and the y -intercept is 850 . What do the slope and y -intercept represent in Inez's situation?

The y -intercept indicates that the phone card started with 50 minutes. The slope indicates that 850 minutes were used per day.

The slope indicates that the phone card started with 850 minutes. The y -intercept indicates that 50 minutes were added per day.

The y -intercept indicates that the phone card started with 850 minutes. The slope indicates that 50 minutes were used per day. ✓

The slope indicates that the phone card started with 50 minutes. The y -intercept indicates that 850 minutes were added per day.

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A local public library decides to track the number of hours that a certain computer is being used. The table represents the number of hours, y , which is dependent on the number of days, x .

x	y
3	21
5	37
7	53

What is the linear equation that represents this situation?

$$y = 2x - 32$$

$$y = 8x - 3$$



$$y = 2x + 16$$



$$y = 8x + 45$$

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Jill converted the equation of the line $15x - 4y = -2$ into slope-intercept form and found the slope and y -intercept of the line as follows.

Jill's Work

$$15x - 4y = -2$$

$$15x - 4y - 15x = -2 - 15x$$

$$-4y = -2 - 15x$$

$$\frac{-4y}{-4} = \frac{-2-15x}{-4}$$

$$y = \frac{-2}{-4} - \frac{-15x}{-4}$$

$$y = \frac{1}{2} - \frac{15}{4}x$$

$$y = -\frac{15}{4}x + \frac{1}{2}$$

$$\text{slope} = -\frac{15}{4}$$

$$y\text{-intercept} = \frac{1}{2}$$

What was her mistake?

She mixed up the slope and the y -intercept.



She got the sign of the y -intercept wrong.

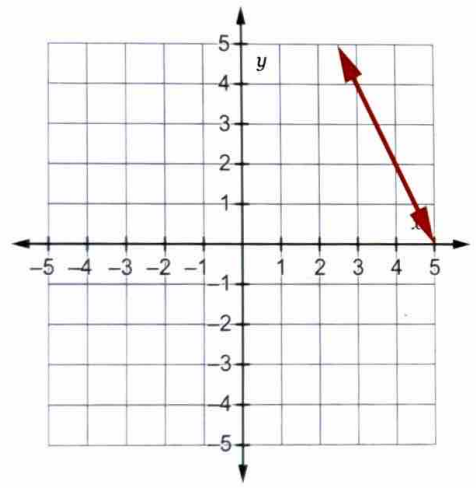
She got the sign of the slope wrong.



She found the reciprocals of the slope and the y -intercept.

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Which linear function has the same y -intercept as the one that is represented by the graph?

x	y
-6	-26
-4	-14
4	34
6	46

x	y
-5	-15
-3	-11
3	1
5	5

x	y
-4	-26
-2	-18
2	-2
4	6

x	y
-3	-4
-1	2
1	8
3	14

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Writing Equations for Linear Relationships

The table represents the total miles traveled, y , after a number of hours, x .

Hours, x	Miles, y
2.5	150
4.0	240
5.5	330
7.0	420

Which linear equation represents the situation?

$$y = 60x$$



$$y = 270x$$

$$y = 4x + 240$$

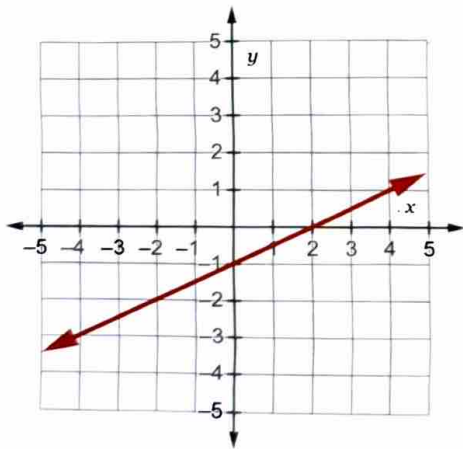
$$y = 60x + 480$$

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Writing Equations for Linear Relationships

Jeremy claims that if a linear function has a slope of the same steepness and the same y -intercept as the linear function in the graph, then it must be the same function.



Which equation proves Jeremy's argument is incorrect?

$$y = -\frac{1}{2}x - 1$$

$$y = -\frac{1}{2}x + 1$$

$$y = \frac{1}{2}x - 1$$



$$y = \frac{1}{2}x + 1$$

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the cost for each hour it takes to repair the machine

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Writing Equations for Linear Relationships



What is the equation of the line passing through the points $(2, -1)$ and $(5, -10)$ in slope-intercept form?

$$y = -3x - 5$$

$$y = 3x - 5$$

$$y = -3x + 5$$



$$y = 3x + 5$$

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Writing Equations for Linear Relationships



The equation of the line that passes through points $(0, -7)$ and $(2, -1)$ is shown. What value is missing from the equation?

$$y = A \boxed{3} x - 7$$

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Writing Equations for Linear Relationships



Consider the linear function that is represented by the equation $y = -10x + 6$ and the linear function that is represented by the equation $y - 36 = 8(x - 4)$. Which statement is correct regarding their slopes and y -intercepts?

The function that is represented by the equation $y - 36 = 8(x - 4)$ has a steeper slope and a greater y -intercept.

The function that is represented by the equation $y = -10x + 6$ has a steeper slope, and the function that is represented by the equation $y - 36 = 8(x - 4)$ has a greater y -intercept.

The function that is represented by the equation $y - 36 = 8(x - 4)$ has a steeper slope, and the function that is represented by the equation $y = -10x + 6$ has a greater y -intercept.

The function that is represented by the equation $y = -10x + 6$ has a steeper slope and a greater y -intercept. ✓

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Writing Equations for Linear Relationships



What is the point-slope form of a line that has a slope of $\frac{1}{2}$ and passes through point $(-7, 2)$?

$$y - 7 = \frac{1}{2}(x - 2)$$

$$7 - y = \frac{1}{2}(-2 - x)$$

$$y - 2 = \frac{1}{2}(x - (-7))$$



$$2 - y = \frac{1}{2}(7 - x)$$

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Writing Equations for Linear Relationships

Which is the equation of a line that has a slope of 4 and passes through point (1, 6)?

$$y = 4x + 6$$

$$y = 4x - 3$$

$$y = 4x + 2$$



$$y = 4x - 2$$

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Writing Equations for Linear Relationships



The equation $y - 5 = 6(x + 1)$ is written in point-slope form. What is the equation written in slope-intercept form?

$$y = 6x + 6$$

$$y = 6x - 1$$

$$y = 6x + 11$$



$$y = 6x - 4$$

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Find the slope using the formula $m = \frac{x_2 - x_1}{y_2 - y_1}$, and then substitute one point and the slope into the equation $y = mx + b$ to find the y -intercept.

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What is the equation in slope-intercept form of the linear function represented by the table?

x	y
-6	-18
-1	-8
4	2
9	12

$y = -2x + 6$

$y = 2x + 6$

$y = 2x - 6$ ✓

$y = -2x - 6$

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Writing Equations for Linear Relationships



Which is the equation of a line that has a slope of 1 and passes through point (5, 3)?

$$y = x - 2$$



$$y = x - 5$$

$$y = x + 3$$

$$y = x + 2$$

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Writing Equations for Linear Relationships



What are the slope and the y -intercept of the linear function that is represented by the equation $y = 9x - 2$?

The slope is 9, and the y -intercept is -2 . ✓

The slope is 9, and the y -intercept is 2.

The slope is -2 , and the y -intercept is 9.

The slope is 2, and they y -intercept is 9.

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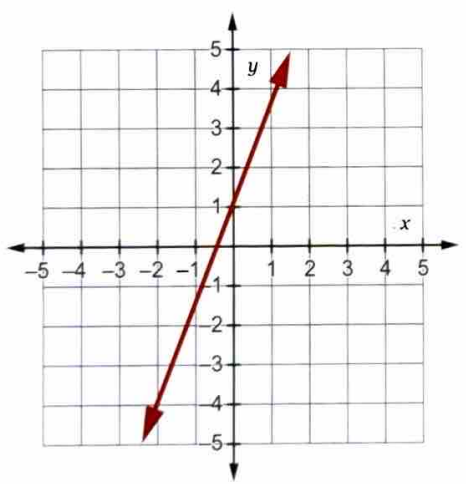
the amount of time that it takes to arrive at the home to make the repairs

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Writing Equations for Linear Relationships



Consider the linear function that is represented by the equation $y = 2x + 2$ and the linear function that is represented by the graph.



Which statement is correct regarding their slopes and y -intercepts?

The function that is represented by the **graph** has a steeper slope, and the function that is represented by the **equation** has a greater y -intercept. ✓

The function that is represented by the **graph** has a steeper slope and a greater y -intercept.

The function that is represented by the **equation** has a steeper slope and a greater y -intercept.

The function that is represented by the **equation** has a steeper slope, and the function that is represented by the **graph** has a greater y -intercept.

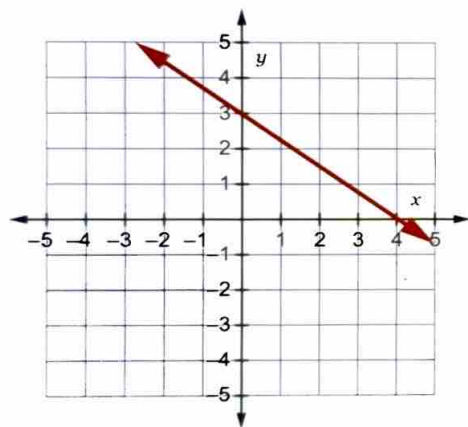
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Writing Equations for Linear Relationships



What are the slope and the y -intercept of the linear function that is represented by the graph?

The slope is $-\frac{3}{4}$, and the y -intercept is 3. ✓

The slope is $-\frac{3}{4}$, and the y -intercept is 4.

The slope is $-\frac{4}{3}$, and the y -intercept is 4.

The slope is $-\frac{4}{3}$, and the y -intercept is 3.

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Writing Equations for Linear Relationships

x	y
$-\frac{3}{4}$	$-\frac{1}{30}$
$-\frac{1}{2}$	$-\frac{2}{15}$
$\frac{1}{4}$	$-\frac{13}{30}$
$\frac{2}{3}$	$-\frac{3}{5}$

What are the slope and the y -intercept of the linear function that is represented by the table?

The slope is $-\frac{2}{5}$, and the y -intercept is $-\frac{1}{3}$.

The slope is $\frac{2}{5}$, and the y -intercept is $-\frac{1}{3}$.

The slope is $-\frac{1}{3}$, and the y -intercept is $-\frac{2}{5}$.

The slope is $\frac{1}{3}$, and the y -intercept is $-\frac{2}{5}$.

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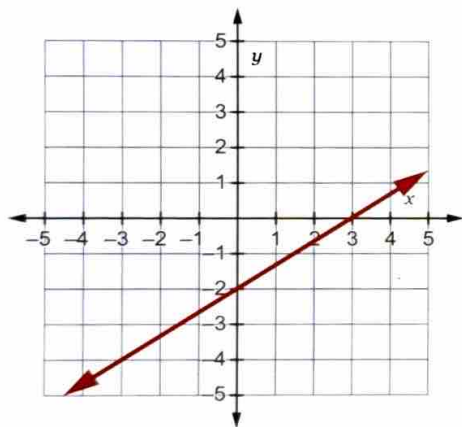
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Writing Equations for Linear Relationships



What are the slope and the y -intercept of the linear function that is represented by the graph?

The slope is $\frac{2}{3}$, and the y -intercept is -2 . ✓

The slope is $\frac{2}{3}$, and the y -intercept is 3 .

The slope is $\frac{3}{2}$, and the y -intercept is 3 .

The slope is $\frac{3}{2}$, and the y -intercept is -2 .

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Writing Equations for Linear Relationships

x	y
$-\frac{1}{2}$	$\frac{1}{5}$
$-\frac{1}{5}$	$\frac{7}{50}$
$\frac{1}{5}$	$\frac{3}{50}$
$\frac{1}{2}$	0

Which linear function has the same slope as the one that is represented by the table?

$$y = -\frac{1}{5}x + \frac{1}{10}$$

$$y = -\frac{1}{2}x + \frac{1}{10}$$

$$y = \frac{1}{2}x - \frac{1}{10}$$

$$y = \frac{1}{5}x - \frac{1}{2}$$

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Writing Equations for Linear Relationships

x	y
-3	18
0	12
3	6
6	0

What are the slope and the y -intercept of the linear function that is represented by the table?

The slope is 2, and the y -intercept is 6.

The slope is -2, and the y -intercept is 6.

The slope is -2, and the y -intercept is 12. ✓

The slope is 2, and the y -intercept is 12.

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Writing Equations for Linear Relationships

x	y
$-\frac{2}{3}$	$-\frac{3}{4}$
$-\frac{1}{6}$	$-\frac{9}{16}$
$\frac{1}{3}$	$-\frac{3}{8}$
$\frac{5}{6}$	$-\frac{3}{16}$

A linear function has an x -intercept of 12 and a slope of $\frac{3}{8}$. How does this function compare to the linear function that is represented by the table?

- It has a different slope and a different y -intercept.
- It has the same slope and a different y -intercept. ✓
- It has the same y -intercept and a different slope.
- It has the same slope and the same y -intercept.

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Consider the linear function that is represented by the equation $y = 4x + 5$ and the linear function that is represented by the table.

x	y
2	16
4	26
6	36
8	46

Which statement is correct regarding their slopes and y -intercepts?

- The function that is represented by the equation has a steeper slope, and the function that is represented by the table has a greater y -intercept.
- The function that is represented by the table has a steeper slope, and the function that is represented by the equation has a greater y -intercept.
- The function that is represented by the table has a steeper slope and a greater y -intercept. ✓
- The function that is represented by the equation has a steeper slope and a greater y -intercept.

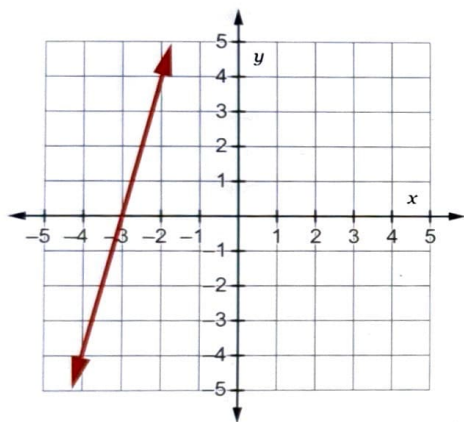
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Writing Equations for Linear Relationships



What are the slope and the y -intercept of the linear function that is represented by the graph?

The slope is 3, and the y -intercept is 12.

The slope is 4, and the y -intercept is 9.

The slope is 4, and the y -intercept is 12. ✓

The slope is 3, and the y -intercept is 9.

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